

LAB EXPERIMENT

ITA0506-COMPUTER VISION

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EXPNO:17 Edge Detection Using Sobel Matrix Along X-Axis

Aim

To detect vertical edges in an image using the Sobel operator along the X-axis using Python and OpenCV.

Theory

The Sobel operator calculates the intensity change in an image.

For X-axis edge detection, the following Sobel matrix is used:

It mainly detects vertical edges in the image.

Algorithm

1. Import cv2 and numpy.
2. Read the input image.
3. Convert the image into grayscale.
4. Apply the Sobel operator along the X-axis.
5. Convert the result to an 8-bit image.
6. Display the original image and Sobel X output.
7. Press any key to close the windows.

CODE:

```
EXP-17.py - C:/Users/POORNA SKHT/OneDrive/Tai ileu/computer vision ita0502/lab experim...
File Edit Format Run Options Window Help
import cv2
import numpy as np

# Read the image
image = cv2.imread("SEN 10.png")

# Check whether image is loaded
if image is None:
    print("Error: Image not found!")
    exit()

# Convert image to grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Apply Sobel operator along X-axis
sobel_x = cv2.Sobel(
    gray,
    cv2.CV_64F,
    1,
    0,
    ksize=3
)

# Convert result to 8-bit
sobel_x = cv2.convertScaleAbs(sobel_x)

# Display original image
cv2.imshow("Original Image", image)

# Display Sobel X edge detection
cv2.imshow("Sobel X Edge Detection", sobel_x)

# Wait for a key
cv2.waitKey(0)

# Close all windows
cv2.destroyAllWindows()
```

OUTPUT :

```
*IDLE Shell 3.13.3*
File Edit Shell Debug Options Window Help
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr 8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-15.py
Homography Matrix obtained using DLT:
[[ 3.18670810e-01 -9.68204638e-01 5.51876644e+01]
 [ 4.03639087e-14 -1.02008128e+00 9.22424098e+01]
 [-2.23790658e-15 -1.19531437e-02 1.00000000e+00]]
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-17.py
```



Ln: 11 Col: 0

Result:

Edge detection was successfully performed using the Sobel operator along the X-axis. The vertical edges present in the image were detected and displayed successfully.